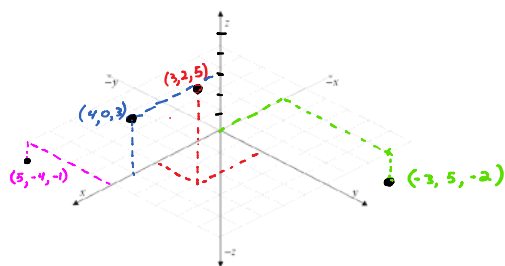


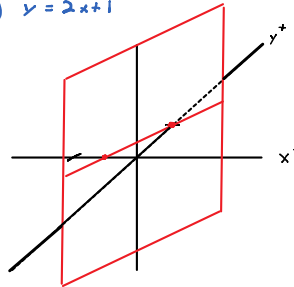
Homework Answers 12.1

①

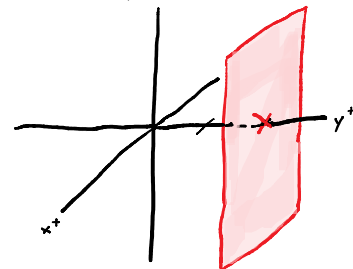


②

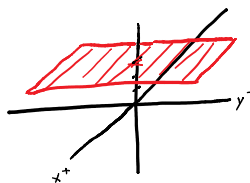
a) $y = 2x + 1$



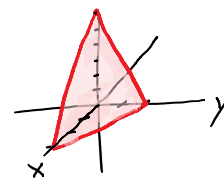
b) $y = 2$



c) $z = 3$



d) $2x + 3y + z = 6$



3.

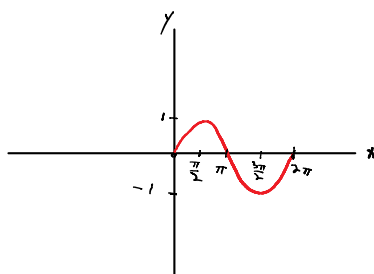
④ $2\sqrt{26}$

- The set of all points in space higher than the horizontal plane $z=5$.
- The set of all points in space between the xy -plane ($z=0$) and the horizontal plane $z=5$, including points on $z=5$.
- The set of all points in space that lie outside the sphere of radius 2 centered at the origin.

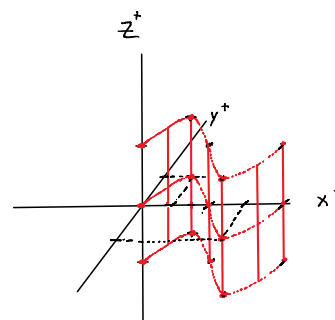
⑤ The intersection is a circle in the yz -plane ($x=0$) with center $(0, 6, -8)$ and radius $= \sqrt{24}$

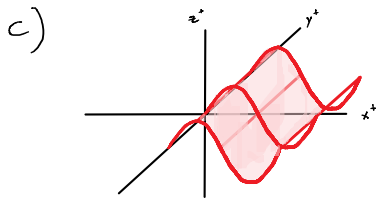
⑥ $(x+7)^2 + (y+5)^2 + (z-2)^2 = 104$

⑦ a)



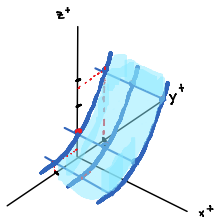
b)





8

a) $z = e^y$



b) $\frac{x^2}{4} + \frac{y^2}{9} = 1$

